

Fig. 2: An actual-size, single-side PCB for stereo audio distribution buffer for headphones

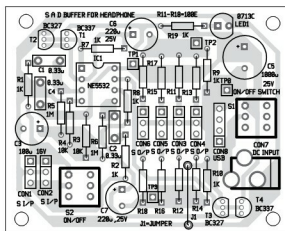


Fig. 3: Component layout for the PCB

the source is present, the amplifier will not work properly in most cases. Also, C1 and C2 act as high-pass filters with the input resistance and should be calculated accordingly as:

$$F_{c1,2} = 1 / (6.28 \times R_{\text{input}} \times C)$$

The input resistance is 1 mega-ohm and depends mainly on the values of R5 and R6. The gain of each channel is unity. Also, in order to keep the circuit simple, potentiometers for volume control are not included.

When switch S2 is closed, both the inputs come in parallel. The outputs on connectors CON3, CON4, CON5 and

CON6 are for headphones or audio interfaces. Any combination of the outputs can be used. The maximum total output current provided by the buffer is around 100 mA.

The buffer itself is built around the dual operational amplifier NE5532 (IC1) and a few discrete components. Other operational amplifiers such as RC4560, OPA2132, TL072, TL082 and even LM358 can be used but it is preferable to use amplifiers for audio which are capable of driving loads of 600 ohms or less and have high input impedance.

The buffer can work with a wide range of power supplies depending on the operational amplifier used. When low-voltage rail-to-rail operational amplifiers are used, the buffer will work from a 5V USB power supply. The circuit will also work with 5V/12V DC wall adaptors providing at least 200mA current.

The quiescent current of the buffer is practically equal to the quiescent current of the operational amplifier. There is no need to put heat-sinks for transistors T1 through T4 but it is preferable to use transistors with a gain of more than 100. T1 through T4 increase the output current of the operational amplifiers. If you remove them, the operational amplifiers will not be able to drive long cables and low-impedance headphones.

Resistors R9 and R10 are used to charge capacitors C6 and C7, respectively, when there is no load. If you remove R9 and R10, you are likely to hear an unpleasant sound of initial

charging of the capacitors. The values of R9 and R10 are not critical (1-100 kilo-ohms) but it is suggested to charge C6 and C7 in less than a second.

The output resistance at CON3 through CON6 is around 100 ohms and can be changed by changing the values of the corresponding series resistors R11 through R18. Usually, the values of these resistors are between 100 and 330 ohms. The outputs have series resistors and connecting or disconnecting a load on one output does not affect noticeably the other outputs.

Working of the circuit is simple. Connect the output from the audio source to CON1. The same audio will be available at ports CON3 through CON6 for four different headphones.

Construction and testing

An actual-size, single-side PCB for stereo audio distribution buffer for headphones is shown in Fig. 2 and its component layout in Fig. 3. After assembling the circuit on the PCB, enclose it in a suitable case. Place stereo input, stereo output and USB connectors on the cabinet and connect them to corresponding locations on PCB using wires.

To test the circuit for proper functioning, check correct power supply at TP1 with respect to TP0. Check audio output for both the channels at TP2 and TP3, respectively. If you are planning to use the buffer with higher power supply voltages, e.g., 30-40 V, it is better to replace transistors BC327/BC337 with transistors similar to BD135/BD136 and appropriate polarised capacitors. ●

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